Back **The Basics of ConvNets**
Graded Assignment • 50 min

Due Mar 30, 11:59 PM CEST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. What do you think applying this filter to a grayscale image will do? 1 / 1 point

$$\begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

- Detecting image contrast.
- Detect vertical edges.
- Detect 45-degree edges.
- Detect horizontal edges.

Correct
Correct. Notice that there is a high delta between the values in the top left part and the ones in the bottom right part. When convolving this filter on a grayscale image, the edges forming a 45-degree angle with the horizontal will be detected.

2. Suppose your input is a 128 by 128 grayscale image, and you are not using a convolutional network. If the first hidden layer has 256 neurons, each one fully connected to the input, how many parameters does this hidden layer have (including the bias parameters)? 1 / 1 point

- 12583168
- 12582912
- 4194304
- 4194560

Correct
Correct, the number of inputs for each unit is 128×128 since the input image is grayscale, so we need $128 \times 128 \times 256$ parameters for the weights and 256 parameters for the bias thus $128 \times 128 \times 256 + 256 = 4194560$.

3. Suppose your input is a 300 by 300 color (RGB) image, and you use a convolutional layer with 100 filters that are each 5x5. How many parameters does this hidden layer have (including the bias parameters)? 1 / 1 point

- 2501
- 7500
- 7600
- 2600

Correct
Correct, you have $25 \times 3 = 75$ weights and 1 bias per filter. Given that you have 100 filters, you get 7,600 parameters for this layer.

4. You have an input volume that is $127 \times 127 \times 16$, and convolve it with 32 filters of 5×5, using a stride of 2 and no padding. What is the output volume? 1 / 1 point

- $62 \times 62 \times 32$
- $123 \times 123 \times 32$
- $123 \times 123 \times 16$
- $62 \times 62 \times 32$

Correct
Correct, using the formula $n_H^{[l]} = \frac{n_H^{[l-1]} + 2 \times p - f}{s} + 1$ with $n_H^{[l-1]} = 127, p = 0, f = 5$, and $s = 2$ we get 62.

5. You have an input volume that is 15x15x8, and pad it using "pad=2". What is the dimension of the resulting volume (after padding)? 1 / 1 point

- 17x17x8
- 19x19x12
- 17x17x10
- 19x19x8

Correct
Correct, padding is applied over the height and the width of the input image. If the padding is two, you add 4 to the height dimension and 4 to the width dimension.

6. You have an input volume that is 63x63x16, and convolve it with 32 filters that are each 7x7, and stride of 1. You want to use a "same" convolution. What is the padding? 1 / 1 point

- 1
- 7
- 2
- 3

Correct
Correct, you need to satisfy the following equation: $n_H - f + 2 \times p + 1 = n_H$ as you want to keep the dimensions between the input volume and the output volume.

7. You have an input volume that is 128x128x12, and apply max pooling with a stride of 4 and a filter size of 4. What is the output volume? 1 / 1 point

- $128 \times 128 \times 3$
- $64 \times 64 \times 12$
- $32 \times 32 \times 3$
- $32 \times 32 \times 12$

Correct
Yes, using the formula $n_H^{[l]} = \frac{n_H^{[l-1]} + 2 \times p - f}{s} + 1$ with $p = 0, f = 4, s = 4$ and $n_H^{[l-1]} = 32$.

8. Which of the following are hyperparameters of the pooling layers? (Choose all that apply) 1 / 1 point

- [ ] $W^{[l]}$ weights.
- [ ] $b^{[l]}$ bias.
- [x] Stride

Correct
Yes, although usually, we set $f = s$ this is one of the hyperparameters of a pooling layer.

- [x] Whether it is max or average.

Correct
Yes, these are the two types of pooling discussed in the lectures, and choosing which to use is considered a hyperparameter.

9. Which of the following are true about convolutional layers? (Check all that apply) 1 / 1 point

- [x] Convolutional layers provide sparsity of connections.

Correct
Yes, this happens since the next activation layer depends only on a small number of activations from the previous layer.

- [x] It allows a feature detector to be used in multiple locations throughout the whole input volume.

Correct
Yes, since convolution involves sliding the filter throughout the whole input volume the feature detector is computed over all the volume.

- [ ] It speeds up the training since we don't need to compute the gradient for convolutional layers.

10. In lecture we talked about "sparsity of connections" as a benefit of using convolutional layers. What does this mean? 1 / 1 point

- Regularization causes gradient descent to set many of the parameters to zero.
- Each filter is connected to every channel in the previous layer.
- Each layer in a convolutional network is connected only to two other layers
- Each activation in the next layer depends on only a small number of activations from the previous layer.

Correct
Yes, each activation of the output volume is computed by multiplying the parameters from **only one filter** with a volumic slice of the input volume and then summing all these together.